

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Eighth Semester of B. Tech (ME) Examination****April-May 2018****ME 408.01/ME 408 Computer Aided Manufacturing****Date: 02.05.2018, Wednesday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below (Each carry one mark).****[07]**

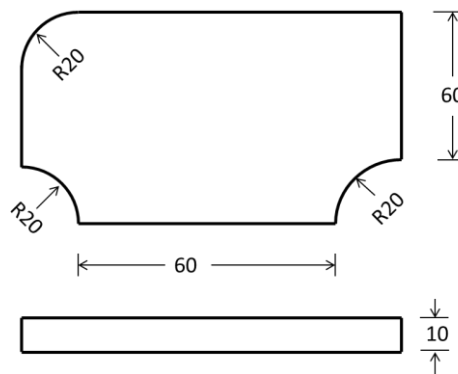
- (i) In a CNC program block, N002 G02 G91 X40 Z40..., G02 AND G91 refer to
 - (a) Circular interpolation in counterclockwise direction and incremental dimension
 - (b) Circular interpolation in counterclockwise direction and absolute dimension
 - (c) Circular interpolation in clockwise direction and incremental dimension
 - (d) Circular interpolation in clockwise direction and absolute dimension
- (ii) The tool of an NC machine has to move along a circular arc from (5, 5) to (10, 10) while performing an operation. The center of the arc is at (10, 5). Which one of the following NC tool path commands performs the above mentioned operation?
 - (a) N010 G02 X5 Y5 R5
 - (b) N010 G03 X5 Y5 R5
 - (c) N010 G01 X10 Y10 R5
 - (d) N010 G02 X10 Y10 R5
- (iii) In finish machining of an island on a casting with CNC milling machine, an end mill with 10 mm diameter is employed. The corner points of the island are represented by (0, 0), (0, 30), (50, 30) and (50, 0). By applying cutter radius right compensation, the trajectory of the cutter will be
 - (a) (-5, 0), (-5, 35), (55, 35), (55,-5), (-5,-5)
 - (b) (0,-5), (55,-5), (55, 35), (-5, 35), (-5,-5)
 - (c) (5, 5), (5, 25), (45, 25), (45, 5), (5, 5)
 - (d) (5, 5), (45, 5), (45, 25), (5, 25), (5, 5)
- (iv) In an NC machining operation, the tool has to be moved from point (5,4) to point (7,2) along a circular path with center at (5,2). The correct G and M code for this motion is
 - (a) N010 G02 X7.0 Y2.0 I0.0 J-2.0
 - (b) N010 G03 X7.0 Y2.0 I5.0 J2.0
 - (c) N010 G01 X7.0 Y2.0 I5.0 J2.0
 - (d) N010 G00 X7.0 Y2.0 I5.0 J2.0
- (v) M30 stands for:
 - (a) End of program
 - (b) End of block
 - (c) End of tape and tape rewind
 - (d) Coolant on/ off
- (vi) Dwell is defined by:
 - (a) G04
 - (b) G03
 - (c) G02
 - (d) G01
- (vii) What do you mean by CL data in CNC programming?

Q – 2.a Explain absolute and incremental modes stating illustration.**[04]**

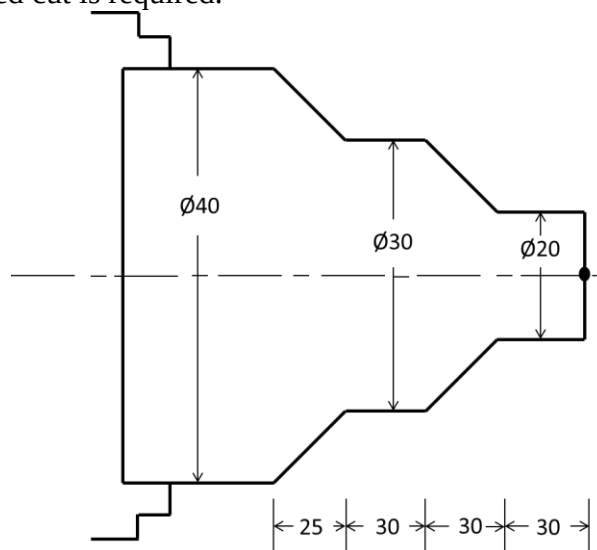
Q – 2.b Answer any three questions.**[12]**

- (i) Explain the terms with respect to CNC machining center with neat sketch:
(i) Machine origin (ii) Program origin (iii) Part origin
- (ii) How the axes are designated in CNC m/c tools? Sketch and designate the axes of CNC lathe machine.
- (iii) Explain what cutter radius compensation is and how it is implemented in CNCs?
- (iv) Differentiate between open loop and closed loop control system.

Q – 3 Write a manual part program for part as shown in figure 1 (All dimensions are in mm) **[12]**
without using cutter radius compensation. Assume tool diameter as 10 mm.

**Figure 1****OR**

Q – 3 Write a manual part program for part shown in figure 2 (All dimensions are in mm). **[12]**
Assume only finished cut is required.

**Figure 2**

SECTION – II

Q - 4 Define following terms (Each carry one mark). [07]

- i. Flexibility in context of FMS
- ii. Automated guided vehicle (AGV)
- iii. Automated storage and retrieval Systems (ASRS)
- iv. Part families
- v. Composite part concept of group technology
- vi. Fused deposition modeling of rapid prototyping
- vii. Computer integrated manufacturing (CIM)

Q – 5.a Explain different types of FMS layouts with neat diagram. [05]

OR

Q – 5.a Explain flexible manufacturing system with diagram. Discuss components of FMS in detail. [05]

Q – 5.b Describe various material handling equipment used in FMS. Enlist types of ASRS. [04]

Q – 5.c Describe rapid prototyping process cycle with the help of suitable diagram. [05]

OR

Q – 5.c Explain selective laser sintering process of RP with sketch. [05]

Q – 6.a Explain the types of coding system structure used in group technology with diagram. [06]

Q – 6.b Describe basic components of robotic system in brief. [04]

Q – 6.c Draw CIM wheel. Explain it in brief. [04]

OR

Q – 6.a Explain MICLASS coding system used in group technology. [05]

Q – 6.b Explain the different elements of CIM in detail. [05]

Q – 6.c Why group technology is more important in the present manufacturing scenario. [04]

G00 Rapid traverse	M00 Program stop	G84 Threading cycle
G01 Linear interpolation	M02 End of program	G90 Absolute mode
G02 Circular interpolation CW	M03 Spindle CW	G91 Incremental mode
G03 Circular interpolation CCW	M04 Spindle CCW	G92 Presetting
G78, G79 Mill cycle	M05 Spindle OFF	M08 Coolant ON
G80 Canned cycle cancel	M06 Tool change	M09 Coolant OFF
		M30 End of tape